

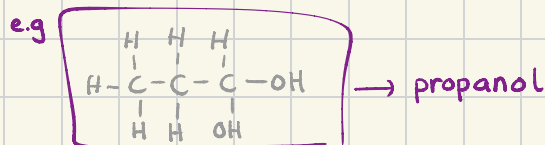
16 - Hydroxy compounds

02/01/25

Hydroxy compounds

→ compounds that contain at least one -OH group

e.g. alcohols



Production of alcohols

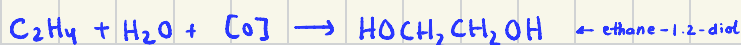
i) electrophilic addition of steam to an alkene

reagent & conditions: $\text{H}_2\text{O (g)}$, H_3PO_4 [catalyst]



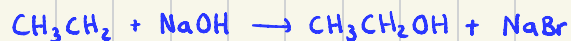
ii) oxidation of alkenes with cold dilute acidified KMnO_4

forms: diol



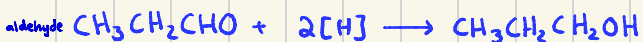
iii) nucleophilic substitution of a halogenoalkane

reagent & conditions: NaOH (aq) , heat



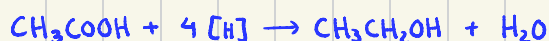
iv) reduction of an aldehyde or ketone

reagent & conditions: NaBH_4 or LiAlH_4



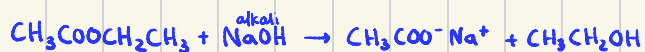
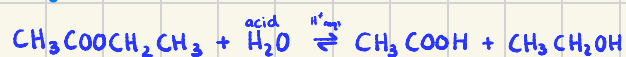
v) reduction of a carboxylic acid

reagent & conditions: LiAlH_4 (or H_2/Ni & heat)

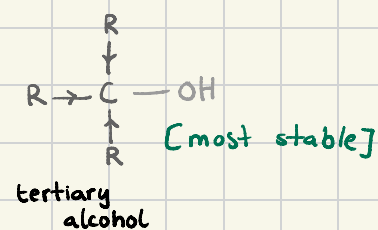
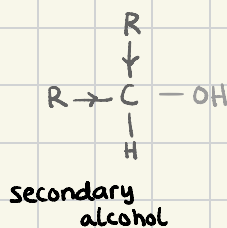
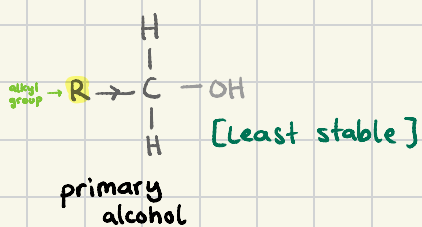


vi) hydrolysis of an ester

reagent & conditions: dilute acid or dilute alkali & heat



Types of alcohols



Identifying alcohols

• mild oxidation with acidified $\text{K}_2\text{Cr}_2\text{O}_7$

primary & secondary alcohols : orange $\rightarrow$ green color change

tertiary alcohols : no color change

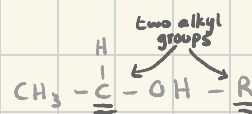
• reaction with alkaline I_2 (aq)

∴ identifies the presence of a $\text{CH}_3\text{CH}(\text{OH})-$ group in an alcohol $\leftarrow$ a secondary alcohol

primary alcohols : no reaction

secondary alcohols : formation of a yellow ppt. of tri-iodomethane + an ion $[\text{RCO}_2^-]$

tertiary alcohols : no reaction



Acidity of alcohols

Acidity is how easily something can lose a hydrogen ion (H^+)

$\Rightarrow$ alcohols & water both have a hydroxyl group BUT alcohols additionally have a carbon chain as well

$\Rightarrow$ the carbon chain has extra electrons that it pushes towards the oxygen

$\Rightarrow$ this makes the oxygen less likely to let go of the hydrogen and as a result :

! alcohols are less acidic than water

$\hookrightarrow$ carbon chain pushes electrons towards the oxygen

$\hookrightarrow$ OH bond becomes stronger

$\hookrightarrow$ harder for alcohols to lose the H^+ (thus its less acidic than water)

if you are wondering ...

only the hydrogen in the $-\text{OH}$ bond can dissociate, this is because the other hydrogens that are attached to the carbons, are nonpolar. This means that they form strong & stable bonds with carbon so those hydrogens don't break off as H^+

Reactions of alcohols

i) combustion (reaction with oxygen)

alcohol + oxygen $\rightarrow$ carbon dioxide + water

ii) substitution to form a halogenoalkane

Reaction with $\overset{\text{halogen}}{HX(g)} \quad \therefore \overset{\text{alcohol}}{CH_3CH_2OH} + HCl \rightarrow \overset{\text{halogenoalkane}}{CH_3CH_2Cl} + H_2O$

Reaction with KCl & conc. $H_2SO_4/H_3PO_4 \quad \therefore CH_3CH_2OH + KCl \xrightarrow[\text{H}_2SO_4]{\text{conc.}} CH_3CH_2Cl + H_2O + KHSO_4$

Reaction with PCl_3 or $PCl_5 \quad \therefore 3CH_3CH_2OH + PCl_3 \rightarrow 3CH_3CH_2Cl + H_3PO_4$
 $\therefore CH_3CH_2OH + PCl_5 \rightarrow CH_3CH_2Cl + POCl_3 + HCl$

Reaction with $SOCl_2 \quad \therefore CH_3CH_2OH + SOCl_2 \rightarrow CH_3CH_2Cl + SO_2 + HCl$

iii) reaction with Na(s)

alcohol + sodium $\rightarrow$ sodium alkoxide + hydrogen

eg. $2C_2H_5OH + 2Na \rightarrow 2C_2H_5O^-Na^+ + H_2$
= sodium ethoxide

iv) oxidation to carbonyl or carboxylic compounds

$\Rightarrow$ reagents & conditions: acidified $K_2Cr_2O_7/KMnO_4$

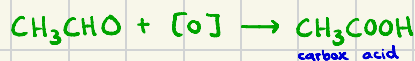
* primary alcohols can be oxidised to aldehydes

—— further oxidation can result in carboxylic acids

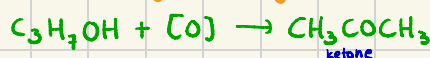
* secondary alcohols can be oxidised to form ketones

* tertiary alcohols cannot be oxidised

oxidation of primary alcohol



oxidation of secondary alcohol



$\Rightarrow$ carbonyl compounds are compounds that have a $C=O$ bond

e.g. aldehydes or ketones

iv) dehydration of alcohols

reagents & conditions: heated catalyst (Al_2O_3) or a conc. acid (H_3PO_4 or P_4O_{10} etc.)

* when alcohol vapour is passed over the hot catalyst or acid, a water molecule is eliminated (dehydration)

alcohol $\rightarrow$ alkene + water

v) alcohol + carboxylic acid (esterification)

reagents & conditions: concentrated H_2SO_4

* alcohol + carboxylic acid $\rightleftharpoons$ ester + water

